1. Clinical Question & Core Hypothesis

Clinical Question

Tumor budding (TBS) at the invasive front is reproducibly visible on H&E and associates with worse outcomes in MIBC, but it is unclear why TBS provides prognostic information beyond conventional stage/grade — specifically, what biological processes make budding prognostically adverse. 2) Tier 1 path used: the single provided path (H&E → Tumor budding score → Prognostic outcomes). I built directly on this because it is the only path and it captures the morphological-to-clinical axis. 3) Tier 2 supporting evidence (conceptual, not supplied as formal tiers here): published mechanistic links between budding-like morphology and epithelial-to-mesenchymal transition (EMT), single-cell invasion, and early dissemination support a causal interpretation that budding reflects cells capable of seeding micro-metastases. 4) Why OS as endpoint: overall survival is the most robust, clinically meaningful endpoint for MIBC and best captures the ultimate impact of occult dissemination implied by budding. 5) Deliberate exclusions: no molecular anchor genes were supplied in the single-path input, so I excluded gene-level claims to avoid inventing unsupported molecular features; complementary image features (e.g., TSR or immune infiltrates) and specific EMT transcription factors were also excluded from the core hypothesis to keep focus on the morphological mechanism and a single endpoint?

Working Hypothesis

High tumor budding score (TBS) at the invasive front in muscle-invasive bladder cancer (MIBC) reflects a localized epithelial-to-mesenchymal transition (EMT)-like state in single cells or small cell clusters that detach from the primary tumor and seed occult micro-metastases. This biological state promotes early dissemination, leading to inferior overall survival (OS) independent of traditional clinicopathologic factors. In a rigorously characterized MIBC cohort with standardized TBS assessment, high TBS will independently predict worse OS after adjustment for pathologic stage, tumor grade, lymphovascular invasion, nodal status, and tumor size. The prognostic impact of high TBS will be mediated by measurable occult dissemination at diagnosis (e.g., detectable circulating tumor cells and/or circulating tumor DNA) and by histologic/surrogate EMT features localized to budding cells (e.g., loss of cohesive cell–cell contacts and acquisition of mesenchymal-like morphology in buds). Furthermore, TBS dynamics during neoadjuvant therapy or early postoperative follow-up (persistent or increasing budding) will correlate with therapeutic resistance and poorer outcomes. Integrating TBS with the tumor immune microenvironment (e.g., immune cell infiltration and stromal context) will enhance risk stratification and identify subgroups that may benefit from intensified systemic therapy, adjuvant surveillance, or therapeutic strategies targeting EMT-like biology in budding cells, without requiring predefined molecular anchor genes in the core hypothesis.”,

Element	Value
Imaging feature	Tumor budding score (TBS) at the invasive front
Primary endpoint	OS

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: multivariable Cox models to test independent prognostic value of TBS;

Prediction 2: concurrent measurement of CTCs/ctDNA at diagnosis and after therapy to assess mediation by occult dissemination;

Prediction 3: multiplex immunostaining in bud regions to document partial EMT-like features;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions